

Research Article

GanDTI: A multi-task neural network for drug-target interaction prediction

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ABSTRACT

Drug discovery processes require drug-target interaction (DTI) prediction by virtual screenings with high accuracy. Compared with traditional methods, the deep learning method requires less time and domain expertise, while achieving higher accuracy. However, there is still room for improvement for higher performance with simplified structures. Meanwhile, this field is calling for multi-task models to solve different tasks. Here we report the GanDTI, an end-to-end deep learning model for both interaction classification and binding affinity prediction tasks. This model employs the compound graph and protein sequence data. It only consists of a graph neural network, an attention module and a multiple-layer perceptron, yet outperforms the state-of-the-art methods to predict binding affinity and interaction classification on the DUD-E, human, and bindingDB benchmark datasets. This demonstrates our refined model is highly effective and efficient for DTI prediction and provides a new strategy for performance improvement.

1. Introduction

The drug discovery process needs to study how small compound molecules bound to proteins extensively to screen potential drugs. Normally, the desired drugs have higher binding affinities to the target molecules and lower affinities with the irrelevant ones (Feinberg et al., 2018). Acquiring these experimental data is a time-consuming and laborious task as there are a massive number of drug candidates to screen (Chen et al., 2018). Therefore, computational methods for virtual screening become a promising way to mitigate this problem.

This drug-target interaction (DTI), or compound-protein interaction (CPI) process can not only be denoted as a binary classification problem, but can also be characterized as a regression task that converges to a value, such as half maximal inhibitory concentration (IC₅₀), dissociation constant (K_d), and inhibition constant (K_i) (Öztürk et al., 2018). As a result, previous computational methods for DTI focused on for both classification and regression tasks.

Earlier endeavors used simulation and molecular docking for affinity prediction reached high accuracy with good interpretability (Salsbury, 2010). This branch normally requires high resolution protein 3D structure data. Recent efforts to use deep learning (DL) method for DTI study have quickly gained rising popularity. This powerful method can capture low dimensional feature representations from simple sequence data, such as protein sequence and Simplified Molecular Input Line

Entry System (SMILES) format, in a structure-free way (Kearnes et al., 2016). They can perform in an end-to-end manner without feature engineering (Wan et al., 2019). The success of DL in the fields of image recognition and natural language processing (NLP) was quickly carried over into the bioinformatics field, and the performance was improved by well-defined neural networks. It outperformed traditional machine learning methods such as random-forest (RF) (Liu et al., 2015a), support vector machines (SVM) (Yamanishi et al., 2008) and other similarity rule based approaches (Fokoue et al., 2016).

Convolutional neural network (CNN) is a commonly seen deep learning network, and has been applied for molecule feature learning and interaction modelling in different dimensions. The 1 dimensional convolutional method can be used to encode the protein or compound sequence before further multiple perceptron layer processing (Öztürk et al., 2018; Tsubaki et al., 2019; Li et al., 2020). Meanwhile, since proteins can be transformed to the distance maps, 2D CNN can be utilized to accomplish DTI tasks similar to a visual question answering (VQA) problem (Zheng et al., 2020). Furthermore, with atom spatial coordinate information, 3D CNN is developed to predict the compound protein interaction (Jiménez et al., 2018; Gomes et al., 2017; Stepniewska-Dziubinska et al., 2018). This method requires structural information data which limits its application to specific datasets. Meanwhile, 3D grid representation is inefficient as it corresponds to void spaces in which no atoms reside.

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Since proteins and compounds can also form sequences similar to sentences in an NLP task (Wan et al., 2019), recurrent neural network (RNN) is another frequently used method for protein interaction prediction. It is sometimes combined with CNN as a hybrid network (Zheng et al., 2020; Gao et al., 2018; Karimi et al., 2019). Image recognition and NLP tasks solved by RNN and CNN normally make use of large datasets for high performance. However, protein datasets are relatively succinct if the structural information is not included, so it calls for more efficient methods to deal with the specific data structure.

Then researchers introduced graph neural network (GNN) to the molecule property prediction task (Gao et al., 2018; Lim et al., 2019; Karlov et al., 2020), since molecules are by nature graphs of arbitrary sizes and shapes (Duvenaud et al., 2015), where atoms are vertices and bonds are edges. So it can use the simple encoding of molecular graph to take advantage of the graph structure (Kearnes et al., 2016) and combines the neighboring atoms' features. Therefore, the method is relatively computationally efficient and well suited for relational inferences, and has been explored in some pioneering works. Tsubaki (Tsubaki et al., 2019) used GNN to encode the compounds and used CNN to process the protein vectors, and they are then processed with multiple-layer perceptrons (MLPs) after vector concatenation to classify the binding results. The similar configuration was also adopted in Shuya's multi-object neural network, where the GCN was modified with graph warp unit to achieve high performance in both binding classification and affinity prediction (Li et al., 2020). Jaechang (Lim et al., 2019) utilized the GNN that incorporated 3D structure information and they contrasted the molecules with and without interactions to predict whether binding is active or inactive. Wen (Torng and Altman, 2019) implemented the graph convolutional network in a two step configuration: the unsupervised graph-autoencoder learned the representations of protein pockets, and the graph-CNN extracted the features from both the protein pockets and the ligands. Despite these successes, graph neural network's application in biomolecules related tasks still have more to explore for better performance and interpretability.

To tackle these problems, prior attempts applied the attention mechanism to ligand-protein binding prediction tasks and received noticeable improvement. Attention can aid in visualization of the molecules' binding based on the importance weight score for interpretability. The attention mechanism was initially introduced in the sequence-to-sequence (seq2seq) tasks like machine translations to understand the correlation between the sequence elements (Bahdanau et al., 2014). To date, six major types of attention appeared in NLP and visual tasks: basic attention, multi-dimensional attention, hierarchical attention, self-attention, memory-based attention, and task-specific attention (Hu, 2018). Most of the attentions applied in CPI predictions were based on basic attentions and self-attention. Some were adopted in CNN (Tsubaki et al., 2019), RNN (Zheng et al., 2020, 2018) or GNN's architecture (Lim et al., 2019). The attention of these work maybe in a single direction (Tsubaki et al., 2019), bidirectional (Zheng et al., 2018), and multi-directional (Li et al., 2020). They may also be separate or joint (Karimi et al., 2019). Recent research showed the neural network can dispense the RNN and CNN to obtain high performance and time efficiency (Vaswani et al., 2017) with attention. These latest findings in NLP could potentially contribute to further improvement.

So in this work, we present a simple yet high performance model based on the residual GNN and attention for ligand-protein binding prediction, and we call it GanDTI (GNN-attention network for Drug-Target Interaction). In this network, the compound graph features are learned by the residual GNN and then are integrated in the attention module to formulate the compound-protein vectors for MLP processing. Unlike previous work, this model does not need CNN or LSTM. The network can perform exceedingly well on both binary interaction classification and binding affinity (K_i , IC50) prediction tasks merely utilizing the compound molecular SMILES and protein sequence data in a structure-free and data-efficient manner. Three public datasets are used to test the model and the results show GanDTI outperforms the state-of-

the-art methods on both interaction classification and binding affinity prediction.

2. Methods

2.1. Datasets and data processing

We evaluated the model with (Mysinger et al., 2012; DUD-E (2021)) and human (Liu et al., 2015b) datasets for classification task and the bindingDB inhibition (Gilson et al., 2016; The Binding Database, 2021) dataset for the regression task. These benchmark datasets were evaluated extensively in previous researches for performance comparison.

The DUD-E dataset is introduced as an enhanced version of DUD (a directory of useful decoys). It originally has 22,645 positive examples and 1,407,145 negative examples. We adopt the dataset from the reference (Zheng et al., 2020), where the data is split into three folds. They also keep the target of the same protein family in the same folder. In total, we picked up 91,239 samples of data, in which 45,609 are negative and 45,629 are positive for training. We randomly chose negatives from the original dataset to make the training dataset balanced, and used the unbalanced dataset for testing.

Human dataset was created in the work (Liu et al., 2015b). They used a systematic screening method following a similarity rule. we adopted the dataset from the work (Tsubaki et al., 2019) for comparison. It contains 3369 positive interactions between 1052 unique compounds and 852 unique proteins.

BindingDB is a web-accessible database of measured binding affinity focusing on protein interaction with small, drug-like molecules. The downloaded bindingDB inhibition dataset in total has 83,293 samples. Only part of them have IC50 data and part of them have K_i data. We reserved the data with IC50 and K_i values and trimmed the data with only a range given. For this reason, we end up using a dataset with a total of 15,000 IC50 data and 5000 K_i data to speed up the training.

These datasets are all formed by three parts: the compound data, the protein data, and the interaction value. We can describe the dataset as $D = \{(c_m, p_m, y_m)\}$, where m is dataset sample size. The compound data c_m is in the SMILES string representation. For example, "c1nccc2n1ccc2". We first added hydrogen to the data, and then get the atoms and bonds to formulate molecule fingerprints with a radius of 2. Meanwhile, we extracted the adjacency matrix based on the molecule data. These steps are all accomplished with RDKit. Protein data p_m is initially encoded as amino acid sequence. We split them into overlapping sections for semantic analysis (Dong et al., 2005) so that these sections can be like words in a sentence. For instance, "LSAAQ" can be decomposed as "LSA", "SAA", "AAQ". This technique is previously used in NLP for feature extraction and has been proven highly effective for protein features as well. The third part of the data is the predicted value. It is either 0 or 1 for the interaction classification task and a float value for the affinity binding task. Before training the model, the train test split is performed on all the dataset with the ratio of 4:1 after shuffling.

2.2. Network architecture and theories

2.2.1. Residual graph neural network

The molecules are graph structures where atoms are nodes and bonds are edges. Each node's features include atom types, amount of hydrogen atoms, aromaticity and valence. To identify the connectivity of the nodes, adjacency matrix A is used to describe which atoms are bonded together. To update the message information in nodes to its neighbors, a message passing method is widely adopted as:

$$c_i^{(l+1)} = U\left(c_i^{(l)}, \sum m(c_j^{(l)})\right) \quad (1)$$

where $c_i^{(l)}$ is the compound molecule fingerprint features. It includes the information of node i and its edges at hidden layer l . U represents update

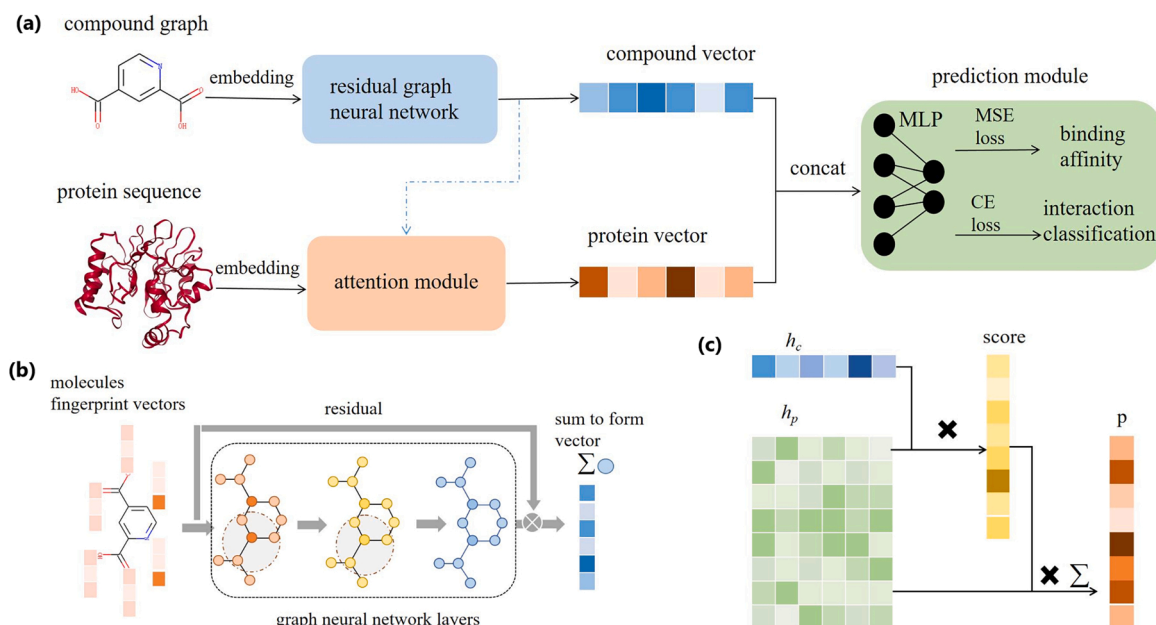


Fig. 1. (a) GanDTT's overall architecture (b) schematic view of the graph neural network's structure to process the compound molecule (c) schematic view of the attention module's matrices and vectors' operation process.

function. m represents the message passing function. The message passing is designed in this way to gather the information on the graph :

$$\sum m = f(W_g A c_i^{(l)}) \quad (2)$$

where W_g is the weight matrix, and f is the non-linear activation function. We use leaky Relu here to improve the performance.

Then after 3 layers of iterations, we added the initial fingerprint features or residual in a feed forward way as:

$$c_i^{(l)} = c_i^{(l)} + c_i^{(1)} \quad (3)$$

where $c_i^{(1)}$ is the first layer's compound molecule fingerprint features. This feature resembles the residual neural network, which is widely adopted in image recognition. Last, the updated fingerprint features are aggregated using summation to form the final compound features:

$$c_e = \sum_{i \in \text{compound}} c_i^{(l)} \quad (4)$$

where c_e is the final aggregated compound vector. This type of aggregation is a frequently used method in GNN. The whole process of GNN is illustrated in Fig. 1(b).

2.2.2. Attention module

The protein data is initially embedded into a sequence representations of $P = (p_1, \dots, p_n)$, where p_i is a vector and n is the embedding dimension. To explore the relationship of the compound vector c and the protein sequence, the attention mechanism is implemented to figure out which part of the protein has larger impact on the interaction. Here the compound vector can be deemed as the key vector and the protein vector as the value vector. The compound and the protein are first transformed to another space in the following way:

$$h_c = f(W_a c_e) \quad (5)$$

$$h_p = f(W_a P) \quad (6)$$

where h_c and h_p are the transformed compound features and protein features, f is the ReLu function, W_a is a learnable weight matrix. We add dot-product attention to them as follows:

$$\text{score}(c_e, P) = \tanh(h_c^* h_p^T) \quad (7)$$

The score value indicates the importance of the protein sequence in the binding process. We used the tanh function to generate the score instead of softmax function, since the long protein sequence could result in too small score values for the model to process. Then based on the score, we can determine the importance of the protein sequence and formulate a summarized protein feature vector as:

$$p_e = \sum_{i \in \text{protein}} W_s \text{score}(c_e, P)^T h_p \quad (8)$$

where W_s is the scale weight of the protein feature. Here we do not use the bidirectional attention since our unidirectional attention shows similar performance. The attention's matrix operation process is illustrated in Fig. 1(c)

2.2.3. Prediction module

After the GNN and attention module, we have the compound vector c_e and the protein feature vector p_e with abundant information for binding affinity prediction. They are then concatenated to generate a vector for MLP processing. The process can be described as:

$$o = \text{MLP}([c_e; p_e]) \quad (9)$$

where o is the output vector, and $[;]$ is concatenation. So the whole problem can simply be described by the function below as:

$$o = g(f(c), h(p, c)) \quad (10)$$

where f is the GNN part, h is the attention module, and g is MLP part.

Our model performs both binding affinity prediction and interaction classification, so the loss functions are defined separately. The object of the binding affinity task is to minimize the mean squared error. It is defined as:

$$L_{MSE} = \frac{1}{n} \sum_{i=1}^n (o_i - y_i)^2 + \frac{\lambda}{2} \|\Theta\|^2 \quad (11)$$

where y_i is the actual measured value and o_i is the predicted value. Θ is the combination of weights and bias in the network, λ is the L2 regularization hyperparameter.

The training object of the classification task is to minimize the cross

Table 1
Evaluation of different methods' performance on the DUD-E Dataset.

Type	Model	AUC	0.5 % RE	1.0 % RE	2.0 % RE	5.0 % RE
ML scoring function	NNscore	0.584	4.166	2.980	2.460	1.891
	RF-score	0.622	5.628	4.274	3.499	2.678
Docking	Vina(Trott and Olson, 2009)	0.716	9.139	7.321	5.811	4.444
	3D CNN(Ragoza et al., 2017a)	0.868	42.559	26.655	19.363	10.710
Deep learning	Pocket GCN(Torng and Altman, 2019)	0.886	44.406	29.748	19.408	10.735
	DrugVQA(Zheng et al., 2020)	0.972	88.17	58.71	35.06	17.39
	GanDTi	0.997	71.13	68.78	49.40	19.79

entropy loss as follows:

$$L_{CE} = - \sum_{i=1}^n (y_i \log(\sigma(o_i)) + (1 - y_i) \log(1 - \sigma(o_i))) + \frac{\lambda}{2} \left\| \theta \right\|_2^2 \quad (12)$$

where σ is the sigmoid function.

2.3. Network implementation

The model was run on a GTX 2080 8GB GPU and implemented with Pytorch 1.5. The Python code for molecule data processing is based on (RDKit (2021)).

The training of the regression tasks needs at least 30 epochs and the classification task needs 15 epoch to complete with a batch size of 1. It takes 20 min to train on the human dataset and 3 h to train on the DUD-E dataset, which is much more time-saving than the peer works. Adam optimizer is used with the learning rate of 0.001 and the weight decay of 10^{-6} . The protein and compound data are transformed by the embedding layer with a dimension of 40. The GNN has three layers as we notice more layers have negligible impact, and two layers of MLP with a dimension of (80, 80) is used before a final layer of (80, 2) for the classification task and (80, 1) for the regression task. We also implement the dropout with a rate of 0.5 before the final layer. All the experiments data provided was repeated at least twice.

2.4. Evaluation metrics

To evaluate the performance of a model for regression task, root mean square error (RMSE) and Pearson correlation coefficient (R) are used. The equation to describe the two metrics are as follows:

$$RMSE(y, p) = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - p_i)^2} \quad (13)$$

$$R(y, p) = \frac{\sum_{i=1}^n (y_i - \bar{y})(p_i - \bar{p})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (p_i - \bar{p})^2}} \quad (14)$$

where y and p represent experimental and predicted affinity values. RMSE measures how far the prediction deviate from the actual values. R measures how much correlation exists between the predicted and the experimental value.

The interaction classification use the area under the receiver operating characteristic curve (AUC), precision and recall to measure the performance. We also applied Receiver Operator Characteristic Curve (ROC) enrichment metric (RE) to the DUD-E dataset. RE describe the the ratio of the true positive rate to the false positive rate (FPR) at a given FPR threshold (Ragoza et al., 2017a).

Table 2
Evaluation of different methods' performance on the human dataset.

Method	AUC	Recall	Precision
KNN	0.86	0.82	0.80
RF	0.90	0.84	0.82
Tsubaki	0.978	0.929	0.938
DeepVQA	0.979	0.961	0.954
GanDTI	0.983	0.933	0.960

Table 3
Evaluation of different methods' performance on the bindingDB K_i dataset.

Model	RMSE	R
DeepAffinity	0.74	0.84
DeepDTA	0.782	0.848
MONN	0.764	0.858
GanDTI	0.721	0.900

3. Results and discussion

3.1. Classification task for interaction prediction

3.1.1. DUD-E dataset

We compared our model's performance on the benchmark dataset, DUD-E, with the previously shown methods to demonstrate the effectiveness of our approach. There are baseline methods, such as machine learning scoring based (Ballester and Mitchell, 2010), docking based (Trott and Olson, 2009), and deep learning based methods(Gomes et al., 2017; Torng and Altman, 2019; Ragoza et al., 2017b). The DrugVQA is the state-of the art methods among them. As shown in Table 1, the GanDTI demonstrated significant improved AUC and RE performance over previous works. The AUC achieved by GanDTI is around 99.7 %, which is 2.5 % higher than DeepVQA. The best RE value at 1% reaches 68.78, at 2% could reach 49.40, and at 5% is 19.79. The precision is around 98.3 % and the recall is as high as 99.3 %. These RE values are several times higher than the traditional methods, and are slightly improved over the DeepVQA. The AUC, recall and precision obtained by GanDTI are the highest among peer works to the authors' knowledge.

3.1.2. Human dataset

To further validate the performance of our model on the binary classification task, we also tested the model on the human dataset, which is much smaller than DUD-E. The following baseline are considered: KNN (k nearest neighbor)(Liu et al., 2015b), RF (random forest)(Liu et al., 2015b), Tsubaki's work(Tsubaki et al., 2019) and DeepVQA (Zheng et al., 2020). We compared the AUC, precision and recall of these models with ours as shown in Table 2. Our results are slightly better than recently emerged models and far superior than the traditional machine learning methods. The AUC score improvement is not as significant as DUD-E, only 0.5 % higher than the state-of the art methods, which is probably due to the fact that the human dataset is much smaller. To validate our thought, we also did experiments to tailor the DUD-E dataset into much smaller sizes, and it also lead to relatively lower performance. For example, when including 10,000 samples of the DUD-E for training and testing, which approximates the human dataset's size, the AUC dropped to 99.1 %. This proves larger datasets perform better with our model for classification tasks.

3.2. Regression task for affinity prediction

In this section, we study the models' performance on the binding affinity prediction task. A perfect model is expected to approximate the prediction to the measured value with little error while forming a linear line with $R = 1$. We separate bindingDB data into two parts based on the measured properties: IC50 and K_i, and tested separately. In Table 3, we

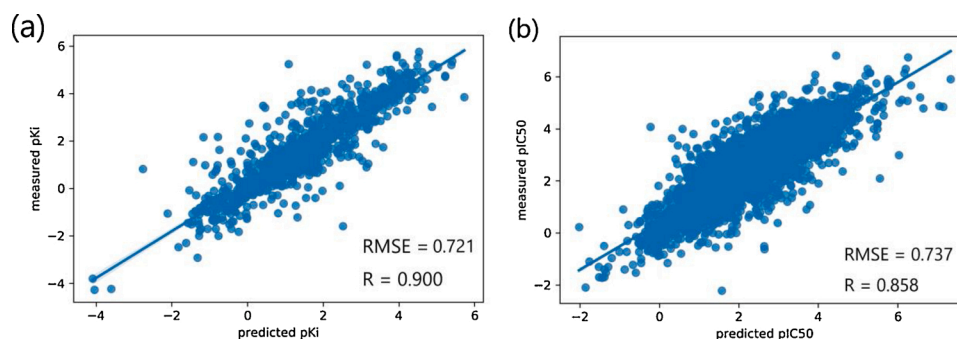


Fig. 2. (a) Affinity prediction of the pKi data versus the measured pKi data (b) Affinity prediction of the pIC50 data versus the measured pIC50 data.

Table 4

Ablation study of GandTI on the datasets.

remove features	Data/Metrics		
	Human/	BindingDB (K_i)/	
	AUC	RMSE	R
Residual	0.980	0.749	0.891
Attention	0.972	0.839	0.861
GNN	0.976	0.736	0.895
Original model	0.983	0.721	0.900

list our model's best performed results and other results from the state-of-the-art models such as DeepAffinity(Karimi et al., 2019), DeepDTA (Öztürk et al., 2018), MONN(Li et al., 2020) on the bindingDB dataset. Our results exceed other methods in both RMSE and R by a large margin. This demonstrate that our simple model could lead to less errors and highly linear predictions on the affinity binding tasks. We show our predicted results versus the measured data of both logarithms of IC50 and K_i data as pIC50 and pKi in Fig. 2. Fig. 2(a) shows the pKi data could reach the RMSE of 0.721 and the R of 0.900, forming a highly linear line. The pIC50 data in Fig. 2(b) is slightly under performing, with the RMSE of 0.737 and the R of 0.858. This may be related to the data's measurement quality or the data amount. These two figures strongly demonstrate our predicted results could align well with the actual measured data with minor errors and the model is well suited for highly accurate binding affinity predictions.

3.3. Ablation study

To further understand our model and analysis which part plays a more important role, we conduct an ablation test on the three datasets. The result is listed in Table 4. We find that the residual, attention and GNN have a little impact on the classification task. The change is only less than 1.1 %, which may be attributed to the task is relative simple for the neural network to solve when the data is preprocessed with our finetuned method. Meanwhile these module's impact on the regression task is more profound. We notice the attention mechanism plays a critical role in improving the performance of the affinity prediction task, since it boosts the performance of the RMSE data from 83.9%–72.1%. This proves the effectiveness of our attention module. The residual and GNN also have some impact but their impact on the RMSE is less than 3.3 % and on the R is around 1%.

4. Conclusion

Accurate virtual drug screenings are crucial to boost discovery throughput and lower the R&D cost. Deep learning shows promising results for this task without requiring domain expertise or structure information in a time-saving manner. Here we present a simple and multi-purpose model, GandTI, that could both predict the binding affinity and

classify the interaction with excellent performance. The model employs a residual graph neural network to process the compound fingerprint data and forms a vector that could project product-based attention on the protein sequence to determine the binding importance on the sequence. Then the two parts of the data are concatenated for MLP processing. The whole process is not implemented alongside the traditionally used techniques like LSTM and CNN. We evaluate the method on three public benchmark datasets and we find that the model outperforms the state-of-the-art deep learning methods in both tasks after comparisons using various metrics. In conclusion, this improved model suggests a new strategy for deep learning based drug virtual screening in a simple and improved way. Future works will focus more on the pharmaceutical industries' emerging real examples of compounds and proteins for drug discovery application.

Availability and implementation

The code and data for this work can be found at: <https://github.com/shuyu-wang/GandTI>.

Author statement

Shuyu Wang forms the research idea and implemented the code to perform the experiment. The majority of the manuscript is drafted by the Shuyu.

Peng Shan actively involved in the research idea formation and helped on the research task implementation. He also helped revised the paper.

Yuliang Zhao helped on the graph creation and provided insightful suggestions on the experiment.

Lei Zuo contributed on the language editing and research idea formation.

Declaration of Competing Interest

The authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.compbiolchem.2021.107476>.

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